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Signature

Note:

- Cross your Registration number(with leading zero). It will be evaluated automatically.
- Sign in the corresponding signature field.

Business Process Technologies and Management

Exam: IN2105 / Endterm

Date: Monday 14th February, 2022

Examiner: Prof. Dr. Stefanie Rinderle-Ma, Dr. Juergen Mangler

Time: 08:00 – 09:30

	P 1	P 2	P 3	P 4				
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Working instructions

- This exam consists of **10 pages** with a total of **4 problems**.
Please make sure now that you received a complete copy of the exam.
- The total amount of achievable credits in this exam is 90 credits.
- The amount of credits per problem roughly reflects the number of minutes you should spend.
- Detaching pages from the exam is prohibited.
- Allowed resources:
 - one **analog dictionary** English ↔ native language
- Subproblems marked by * can be solved without results of previous subproblems.
- **Answers are only accepted if the solution approach is documented.** Give a reason for each answer unless explicitly stated otherwise in the respective subproblem.
- Do not write with red or green colors nor use pencils.
- Physically turn off all electronic devices, put them into your bag and close the bag.

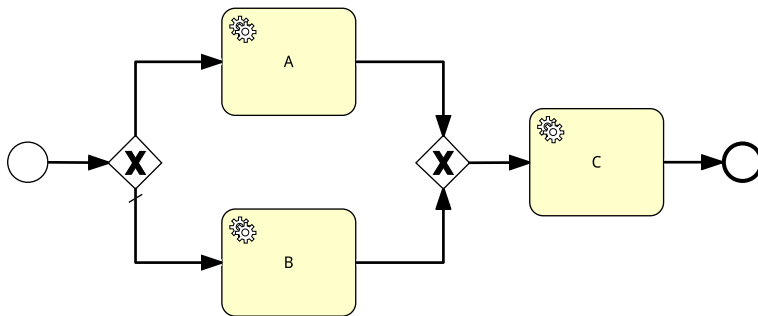
Left room from _____ to _____ / Early submission at _____

Problem 1 Model Translation (24 credits)

Given the following model, write down a pseudo code program, that resembles the model. Please use visible indentation for your pseudo code. Use only the following instructions:

Decision	if CONDITION else if CONDITION else end
Loop	while CONDITION && CONDITION end
Task	task LABEL
Parallel	parallel branch . . . end branch . . . end end
Break out of innermost loop	break
Set variables to VALUE	set VAR = VALUE
Explicit deadlock signaling	deadlock

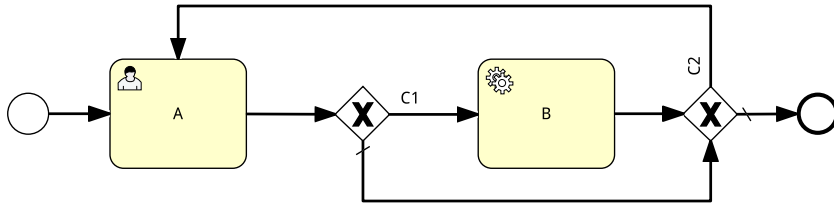
Example:



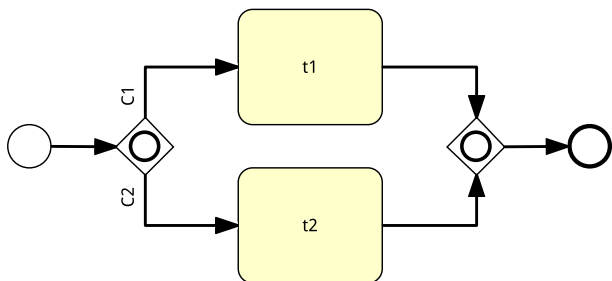
```
if C1
  task A
else
  task B
end
task C
```

Please note, that if no condition for a decision branch is valid, a deadlock occurs. Programming languages work differently, so this has to be accounted for explicitly.

a)* Translate the BPMN to pseudo code. Use only the instructions enumerated above.



b)* Translate the BPMN to pseudo code. Use only the instructions enumerated above.



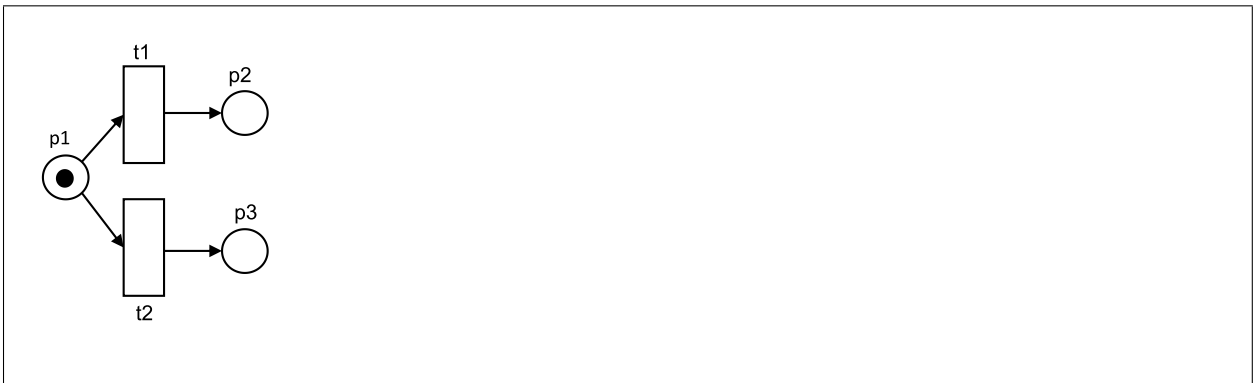
Problem 2 Petri Nets 1 (21 credits)

Please label each transition and each place. Please read the rules for each problem carefully.

a)* Model an inclusive or (decision) that takes place after an initial place p1. The result should be block-structured and include at least places p1...p4. Also draw the equivalent model in BPMN.



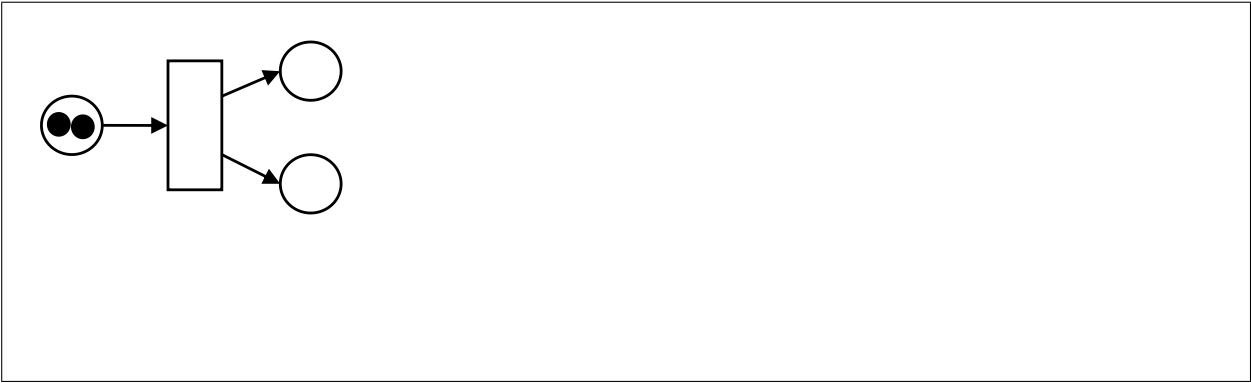
b)* Add a final state to the Petri Net by adding whatever you want, so that an equivalent BPMN would be valid and block-structured. Write the equivalent BPMN down - with the correct labels.



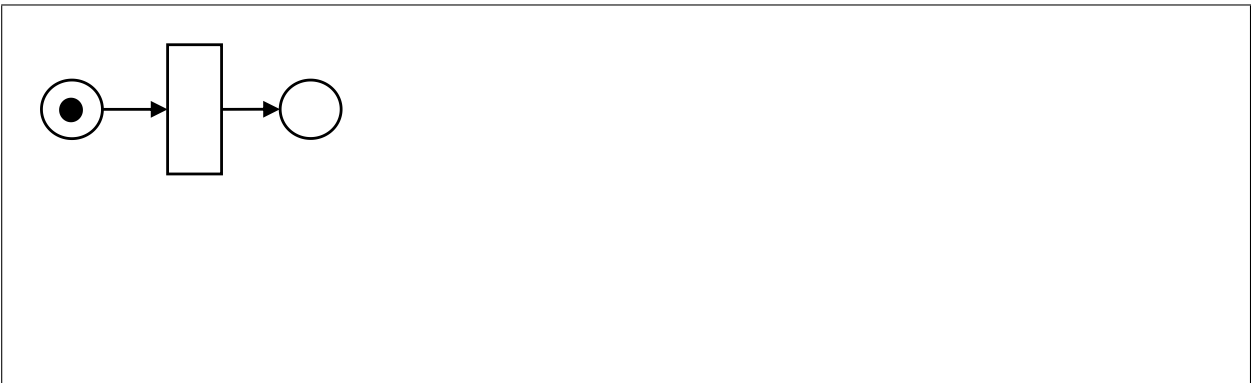
c)* Make the Net bounded and live. Add only transitions and flows. Capacities are not allowed, edge weights only on newly added flows. Please design your solution in a way that at least one marking M_x exists, where 3 tokens are in one place



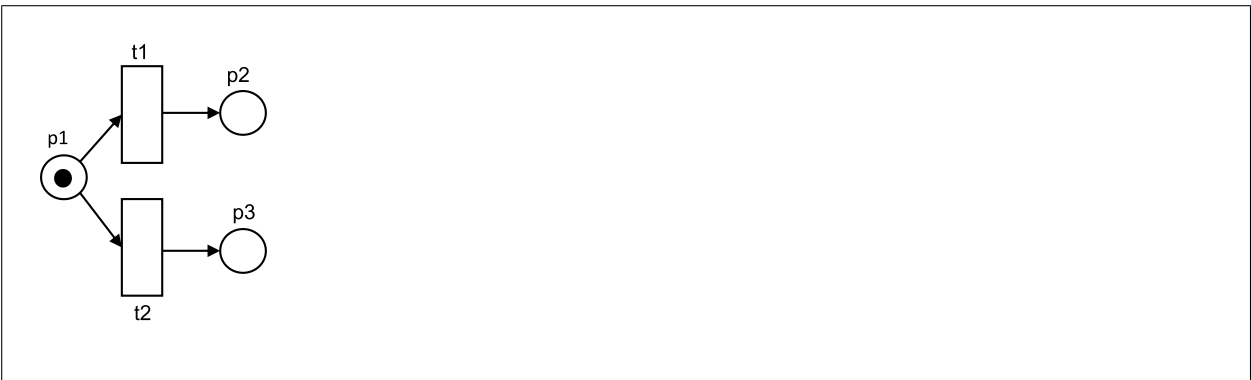
d)* Make the Net live so that ALWAYS two tokens are in the Net. Do not add any new places.



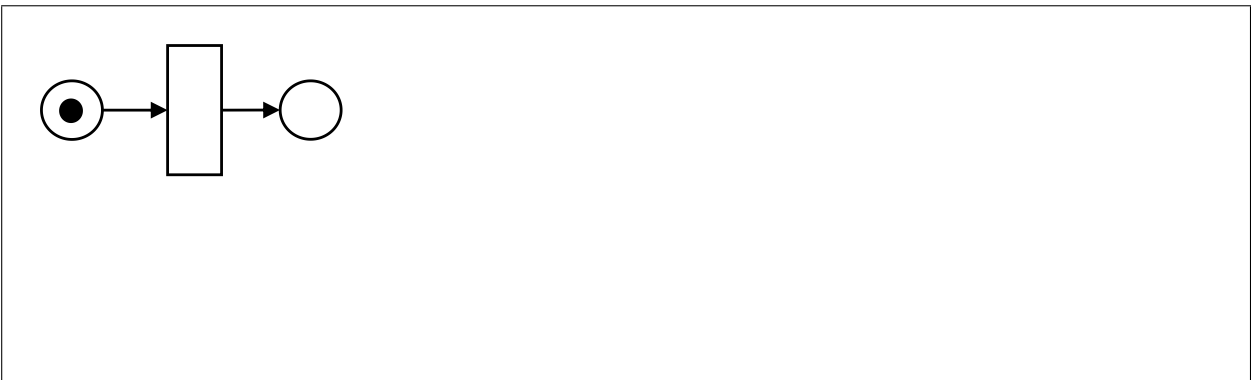
e)* Make the Net not structural sound.



f)* Introduce a deadlock.



g)* Introduce a livelock (without making the Net live).



Problem 3 Petri Nets 2 (24 credits)

Given the following Petri Net:

$$N = (P, T, F, C, W, M_0)$$

$$P = \{p1, p2, p3, p4, p5\}$$

$$T = \{t1, t2, t3, t4\}$$

$$F = \{(p1, t1), (t1, p2), (t1, p3), (t1, p4), (t1, p5), (p3, t3), (p4, t2), (t2, p5), (t3, p5), (p2, t4), (p5, t4), (t4, p1)\}$$

$$C = \{\}$$

$$W = \{(p5, t4, 3)\}$$

$$M_0 = \{(p1, 1)\}$$

a)* Draw the Petri Net!

b)* Perform a Reachability Analysis! Rules: each line should be labelled (marking 0 to marking n). For each transition that fires denote the resulting marking. Please stay within the raster.

c) What kind of Petri Net is this? Why? Describe one other type of Petri Net and its properties.

d) Is the Petri Net live? Which properties of the reachability analysis show this?

e) Is the Petri Net safe? Which properties of the reachability analysis show this?

f) Is the Petri Net bounded? Which properties of the reachability analysis show this?

Problem 4 Comprehension (21 credits)

Read each question carefully. Provide short and precise answers (keywords).

a)* How do humans interact with processes? How can work be distributed? What is the minimal input that this mechanism needs?

b)* What is the purpose of a correlator?

c)* What is the difference between a process choreography and a process orchestration?

d)* Which interaction patterns between a task and its implementation exist?

e)* Describe a typical lifecycle of a process instance.

f)* Describe a typical lifecycle of a process task.

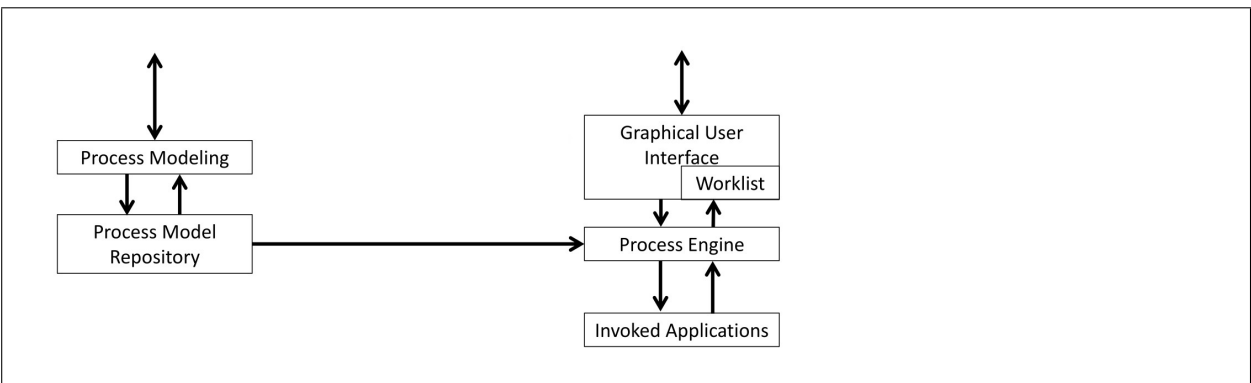
g)* Describe a typical lifecycle of a worklist task.

h)* What is the minimal information an item in a process log has to carry? Give an example for each item.

i)* What is the difference between logging and monitoring?

j)* What are potential retention policies, when a correlator has to store a message?

k)* What are the stakeholders in a process management system?



l)* Why are process logs important?

Additional space for solutions—clearly mark the (sub)problem your answers are related to and strike out invalid solutions.

